

Improving Machine Learning Models for Cosmology with Multimodal Data

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~95% of the universe's mass/energy budget is unknown

(Dark Energy Survey)

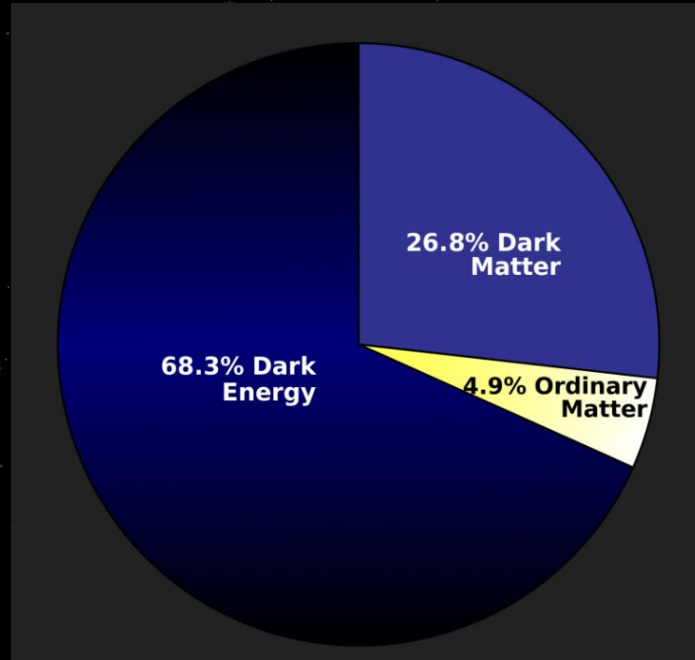


Image credits: Wikipedia

Redshift can help us determine the distance and how fast the universe was expanding at different times.

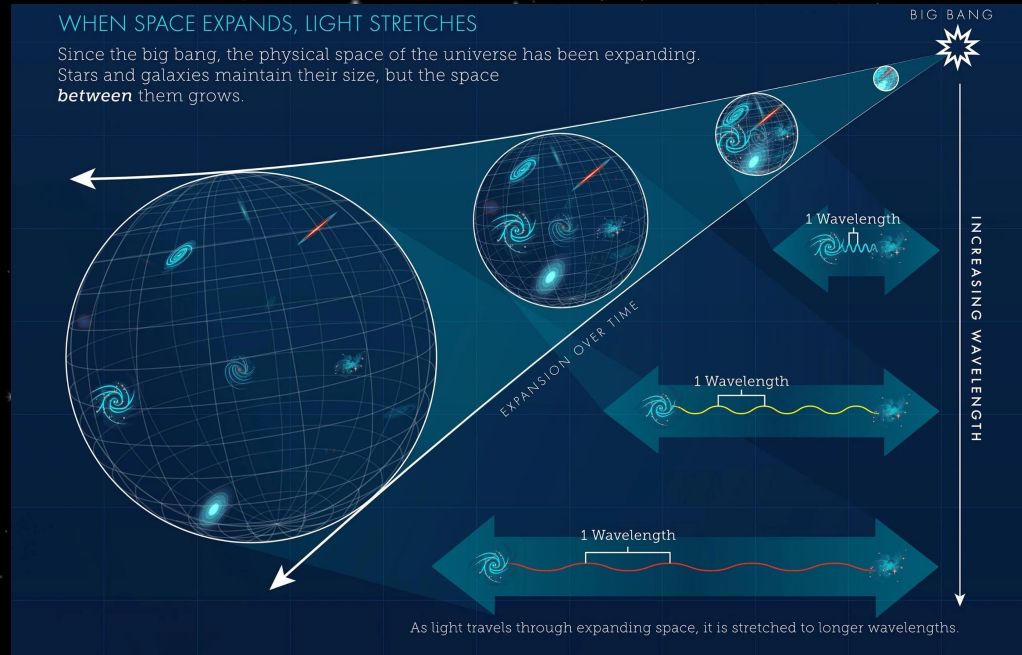


Image credits: NASA

Redshift is measured in two ways

Spectroscopic Redshift

- Involves measuring all wavelengths of well-known spectral lines
- Advantage: accurate
- Disadvantage: time-consuming

Photometric Redshift

- Found by measuring the intensity of several broad bands, used by large surveys
- Advantage: faster
- Disadvantage: needs calibration

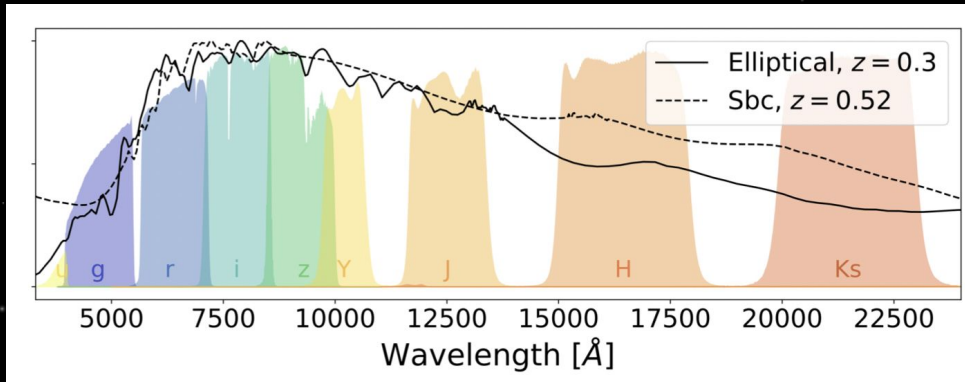


Image credits: Sheldon et al. (2020)

LSST is a wide survey that will image 5 billion galaxies

Large scale surveys, like the Legacy of Space and Time (LSST), will study dark energy by measuring the structure and evolution of the universe.

LSST relies on photometric redshift estimations.

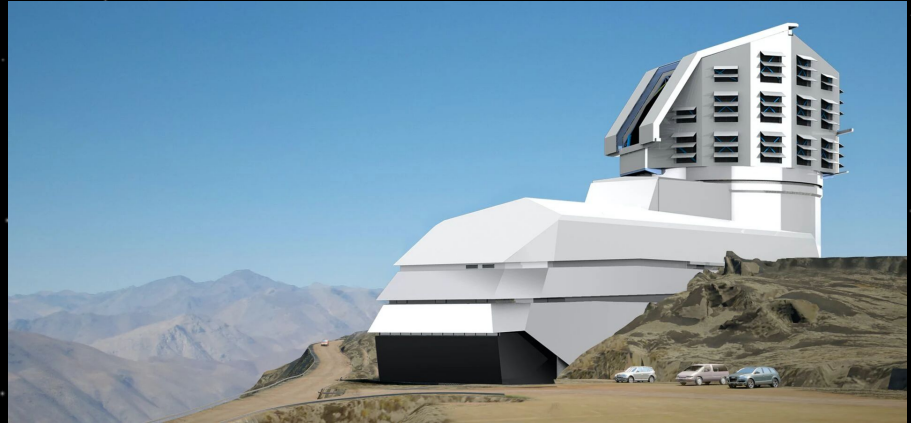
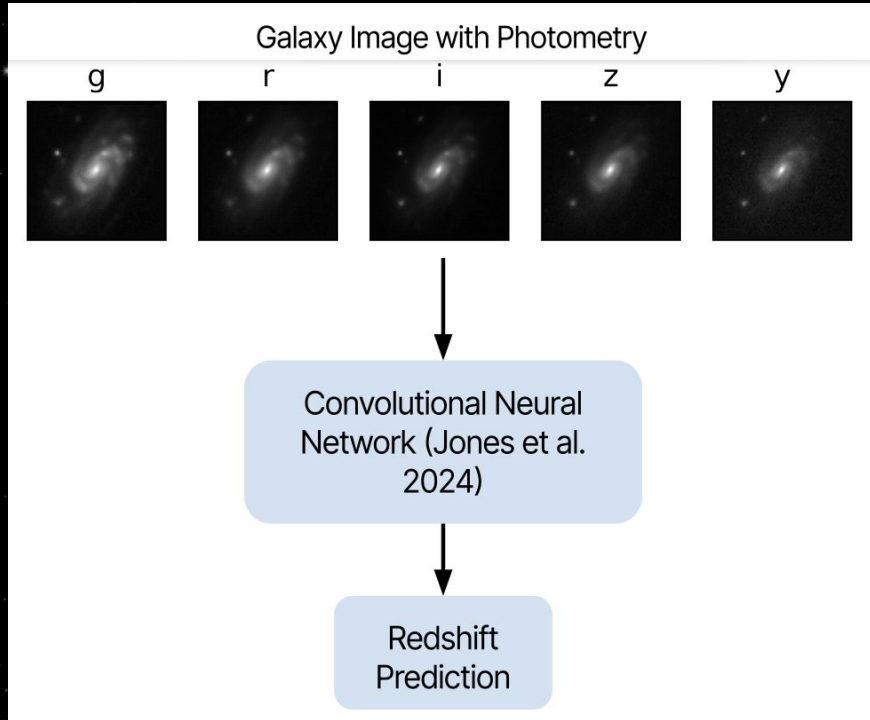


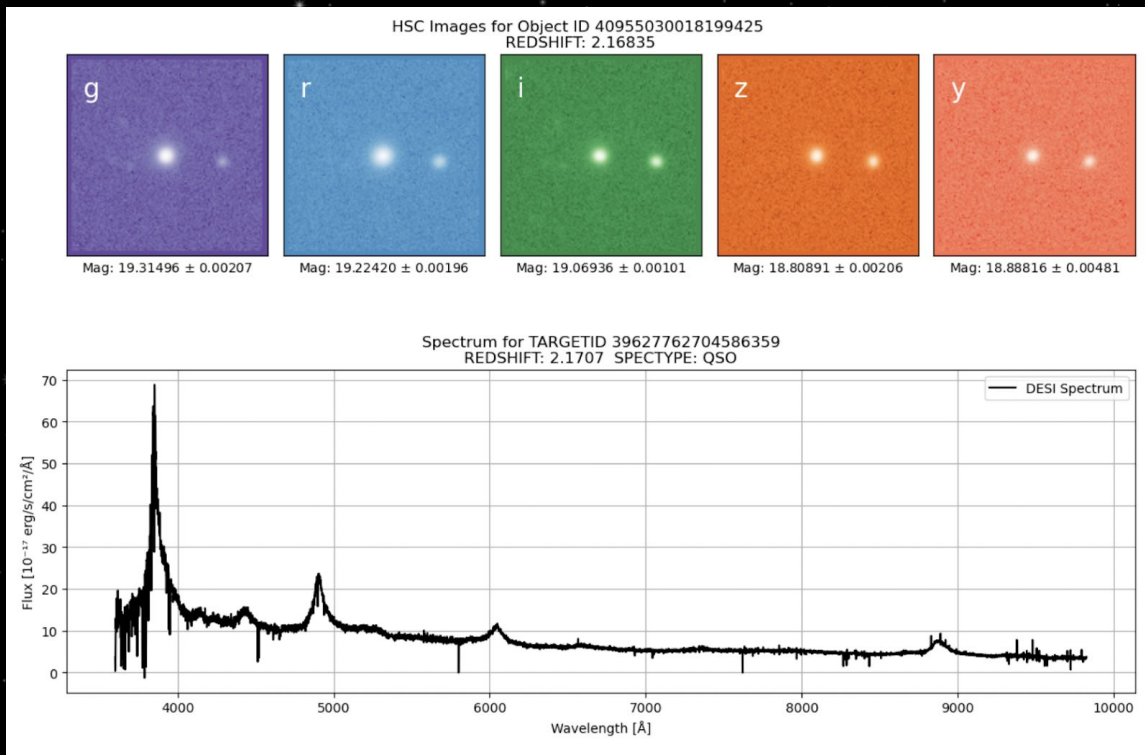
Image credits: Rubin Observatory

Machine learning is used to predict redshift



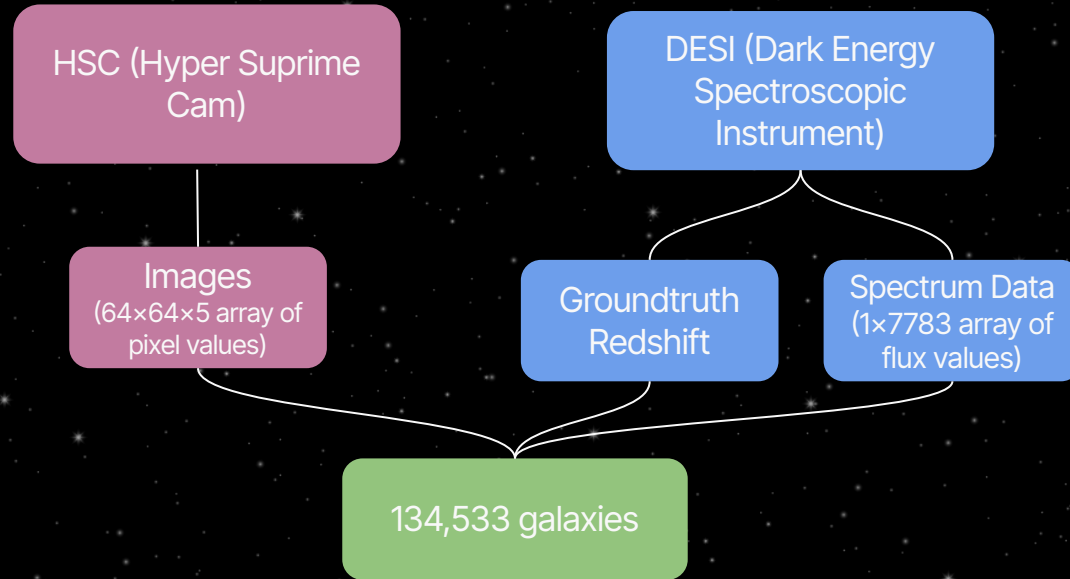
- As the amount of astronomical data grows, we need to process it at scale
- ML can deal with the *volume* and *complexity* of data
- Existing models make redshift predictions from images & photometry
 - Jones et al. 2024 convolutional and probabilistic NNs

Project Goal: Introducing Spectra



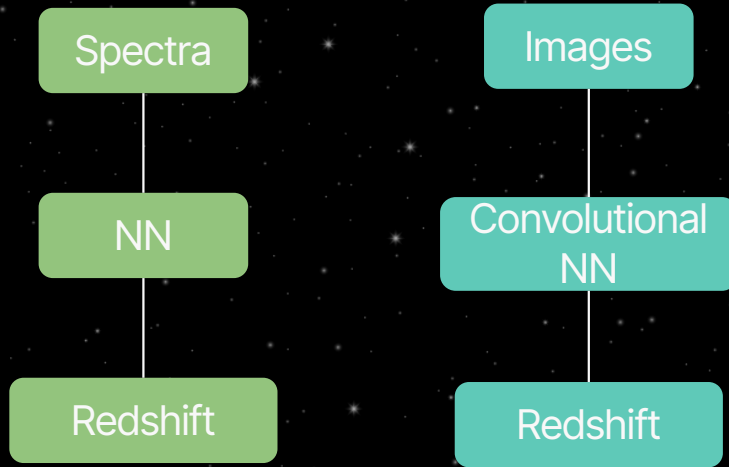
- We want to build on the existing redshift prediction models by adding spectrum data
- Spectra = more physics
 - Difficult to obtain
- How does introducing spectra during training affect the model's performance?

Creating the multimodal dataset of images and spectra

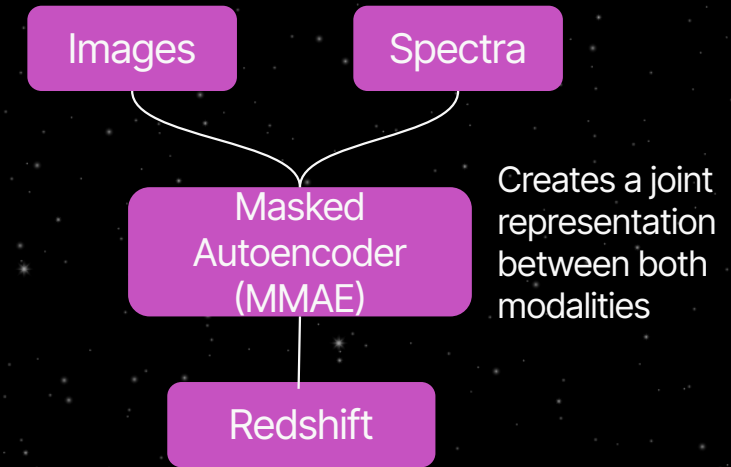


We tested the performance of different neural networks

Traditional Models

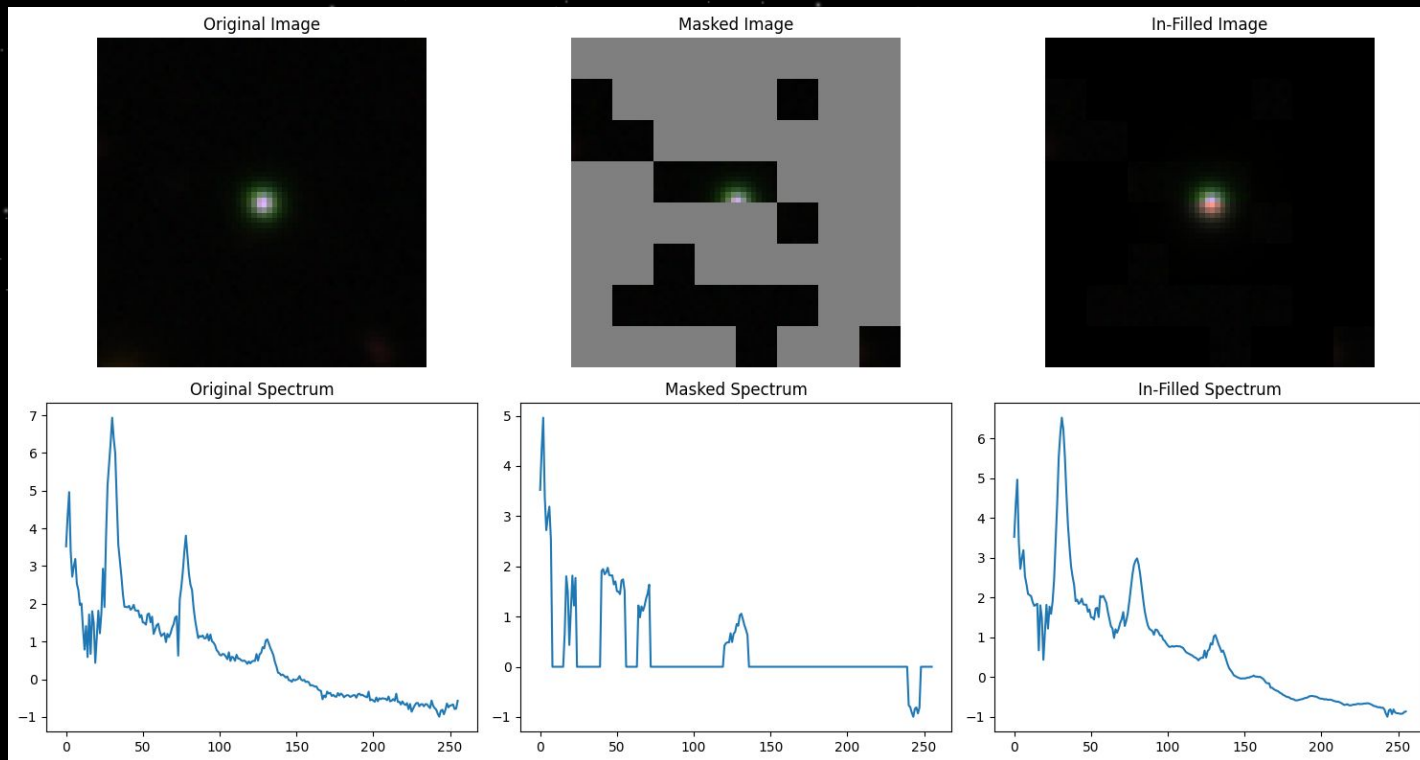


Multimodal



★ Model is evaluated with images only or
images + spectra

Example of Masked Pre-Training Step



Metrics for comparing models

Bias

Average systematic error

$$b = \frac{z_{\text{phot}} - z_{\text{spec}}}{1 + z_{\text{spec}}}$$

Scatter

How spread out the prediction errors are (precision around the true values)

$$\text{Median}(|\Delta z - \text{median}(\Delta z_i)|)$$

RMS

Overall prediction accuracy, combines bias and scatter

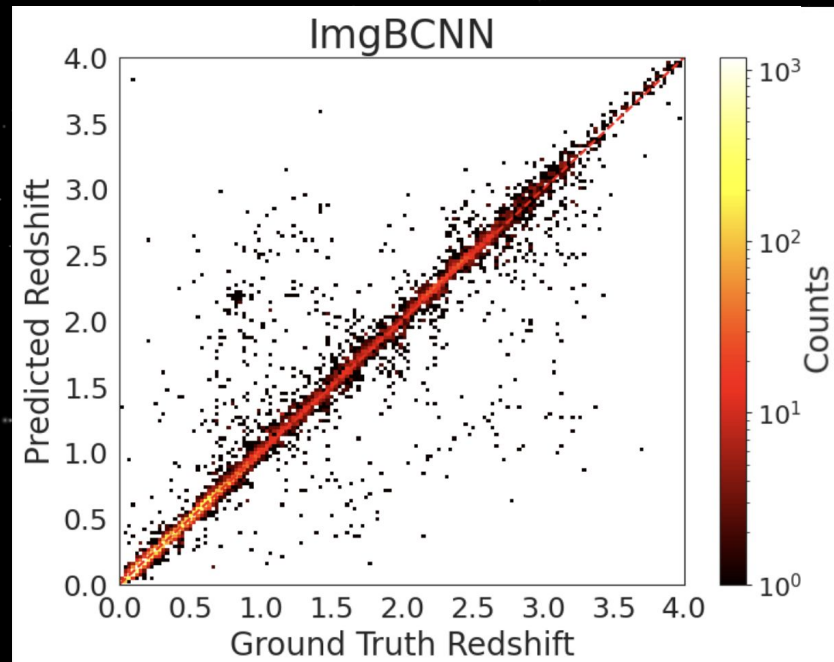
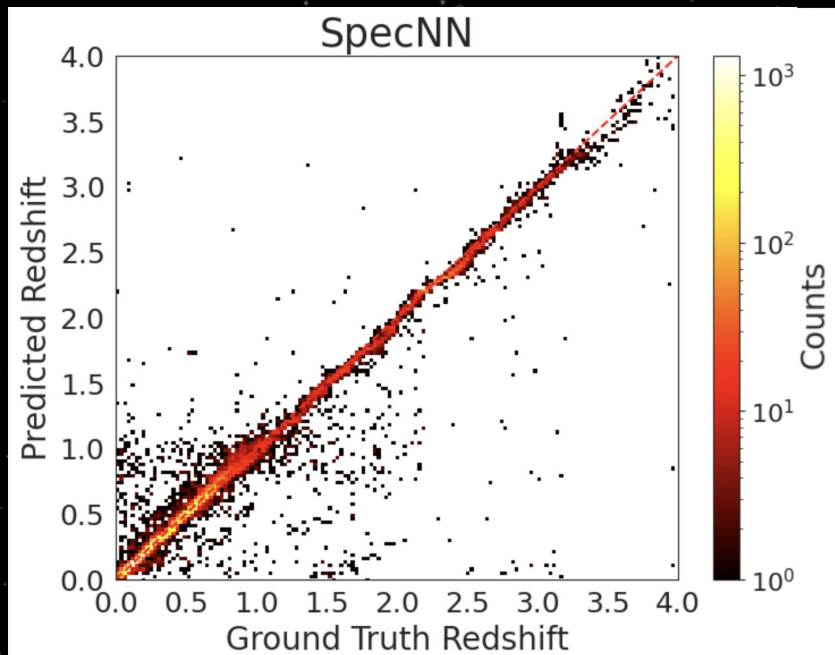
$$\sqrt{\frac{1}{n_{\text{gals}}} \sum_{\text{gals}} \left(\frac{z_{\text{phot}} - z_{\text{spec}}}{1 + z_{\text{spec}}} \right)^2}$$

Outlier Rate

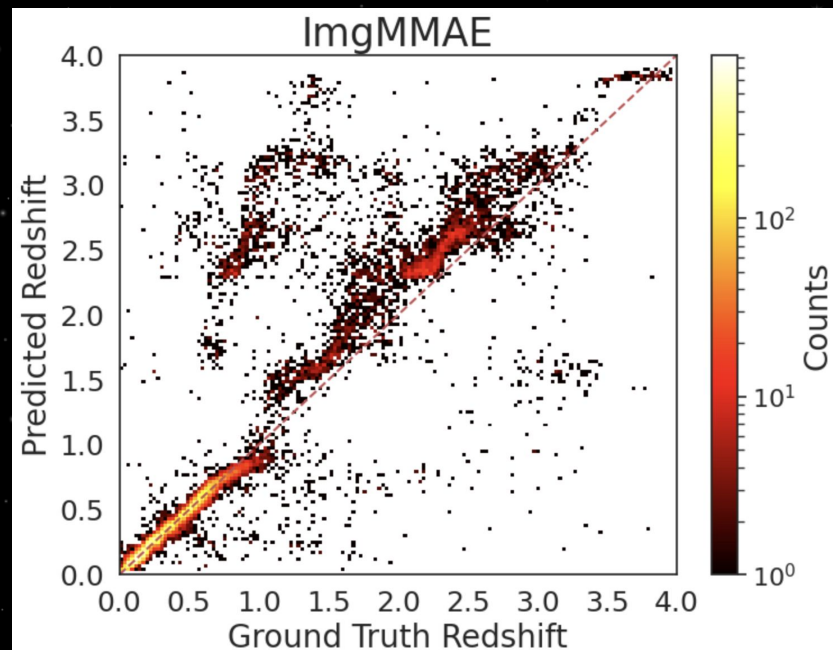
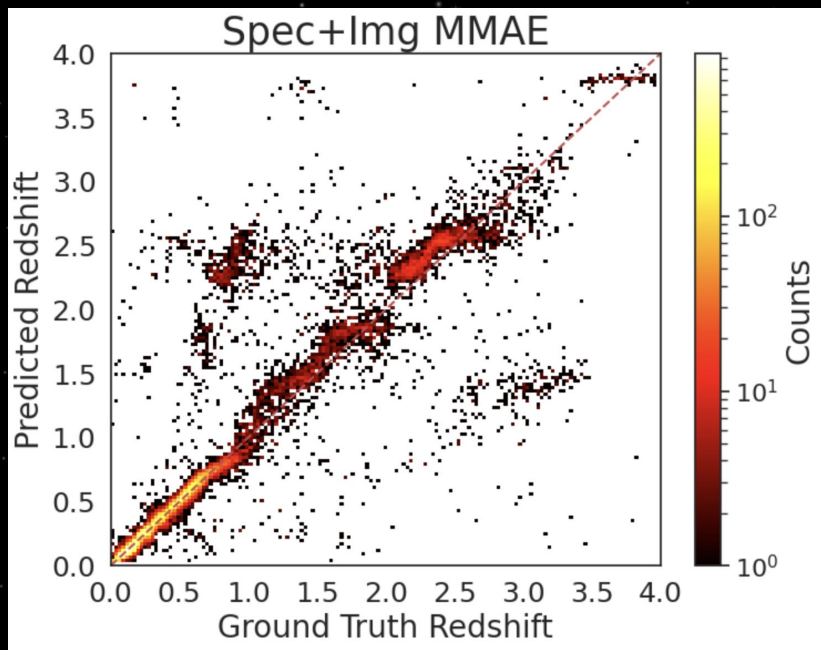
% predictions that are outliers

$$O: \frac{|z_{\text{phot}} - z_{\text{spec}}|}{1 + z_{\text{spec}}} > .15$$

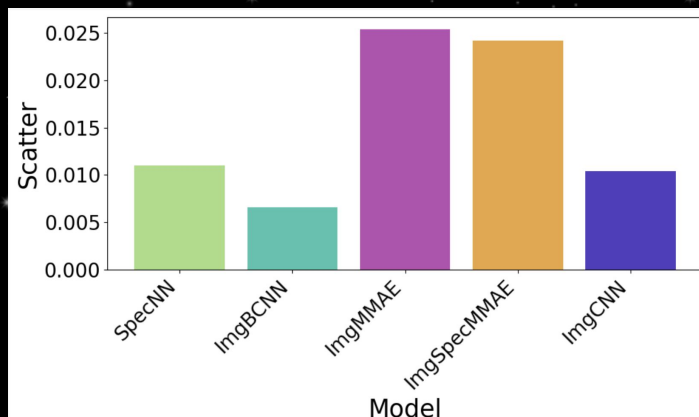
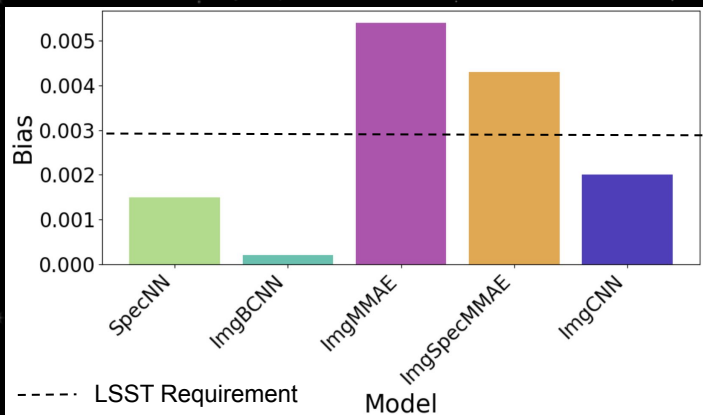
Traditional Model Performance



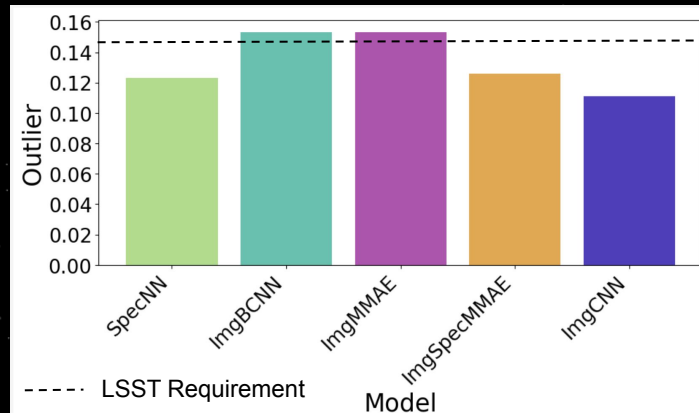
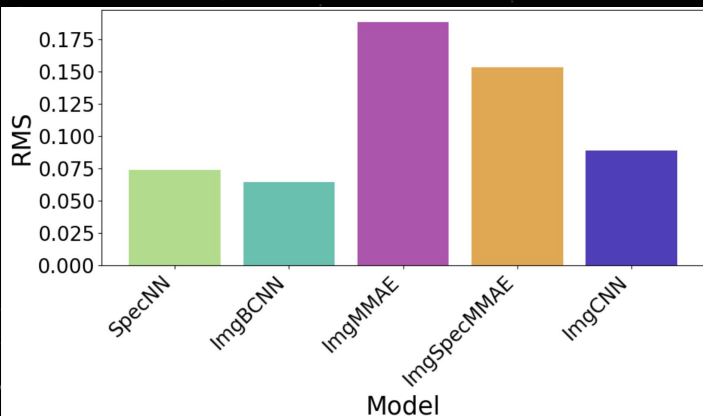
Predictions from the multimodal model



Traditional vs Conditional Performance



Lower =
better!

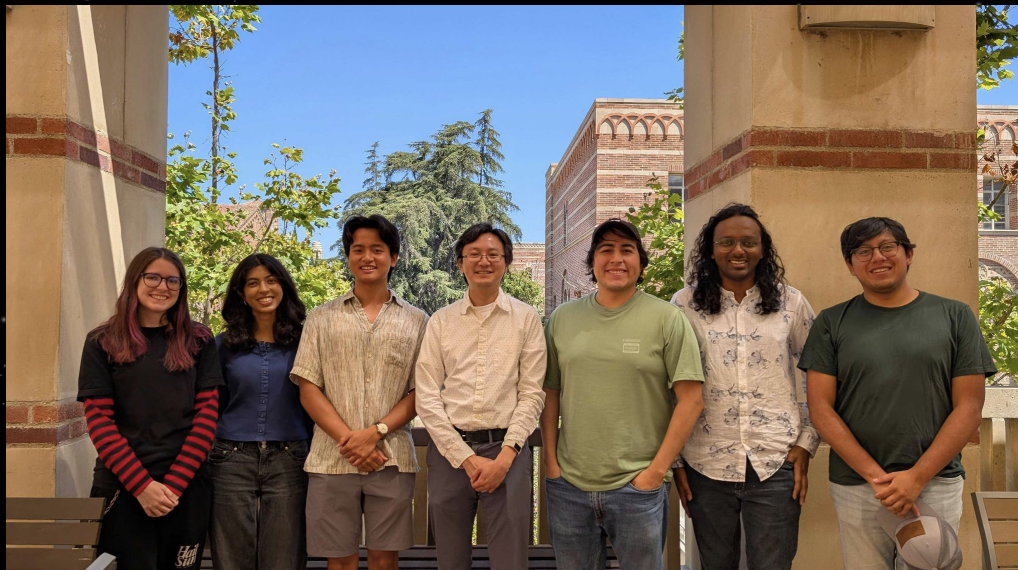


Conclusion and Future Work

- Spectra appears to improve predictions, but MMAE doesn't perform better than the baseline models
- Model struggles when evaluated with images only
 - Need a stronger way of associating modalities
 - Is the model identifying key spectral features?
- Compare to models outside the group and test different architectures

Acknowledgements

- Professor Tuan Do
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- Andrew Lizarraga
- Jonathan Soriano
- Vikram Seenivasan
- Shoko Sakai
- REU donors



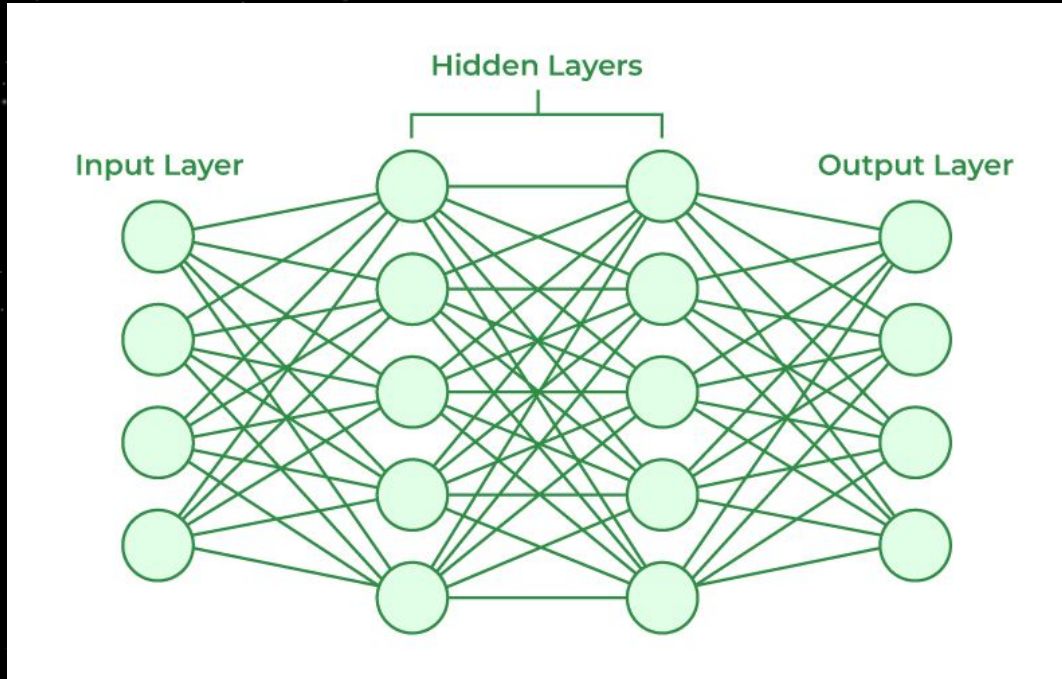


Thank you!

Backup Slides

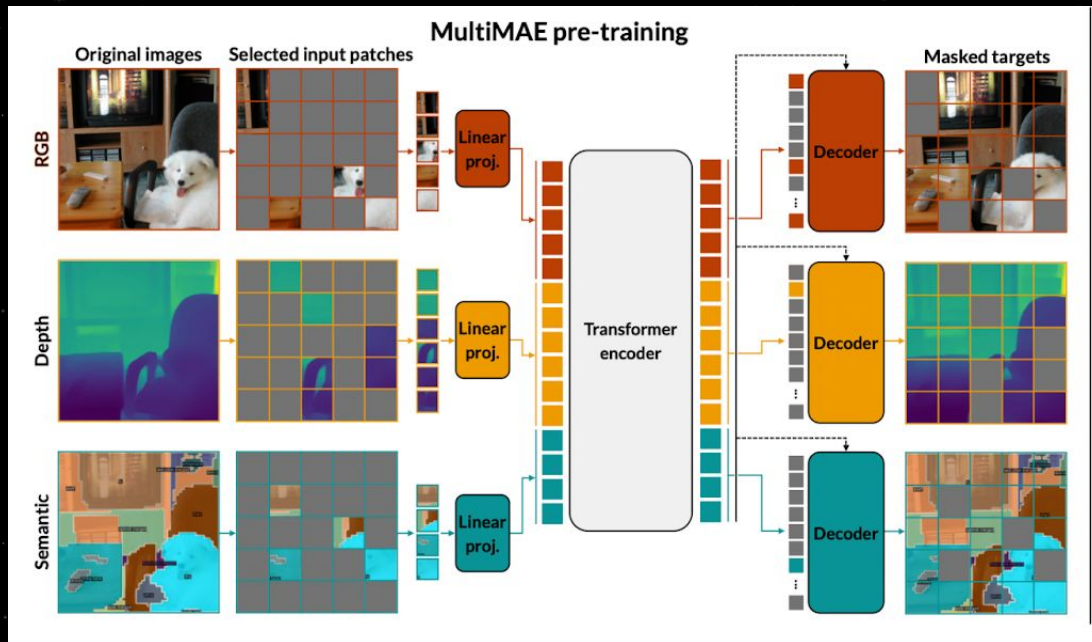
Neural Network

Spectra or
Image Data

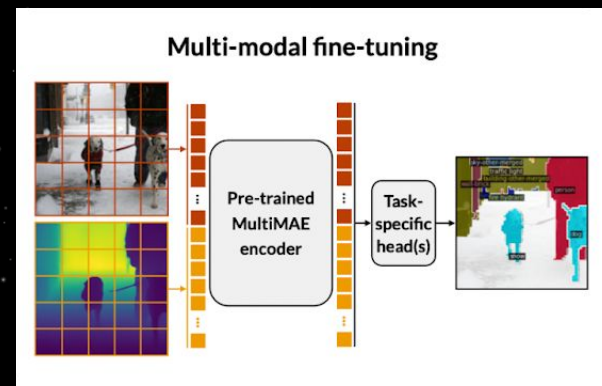


Redshift
Prediction

Masked Autoencoder

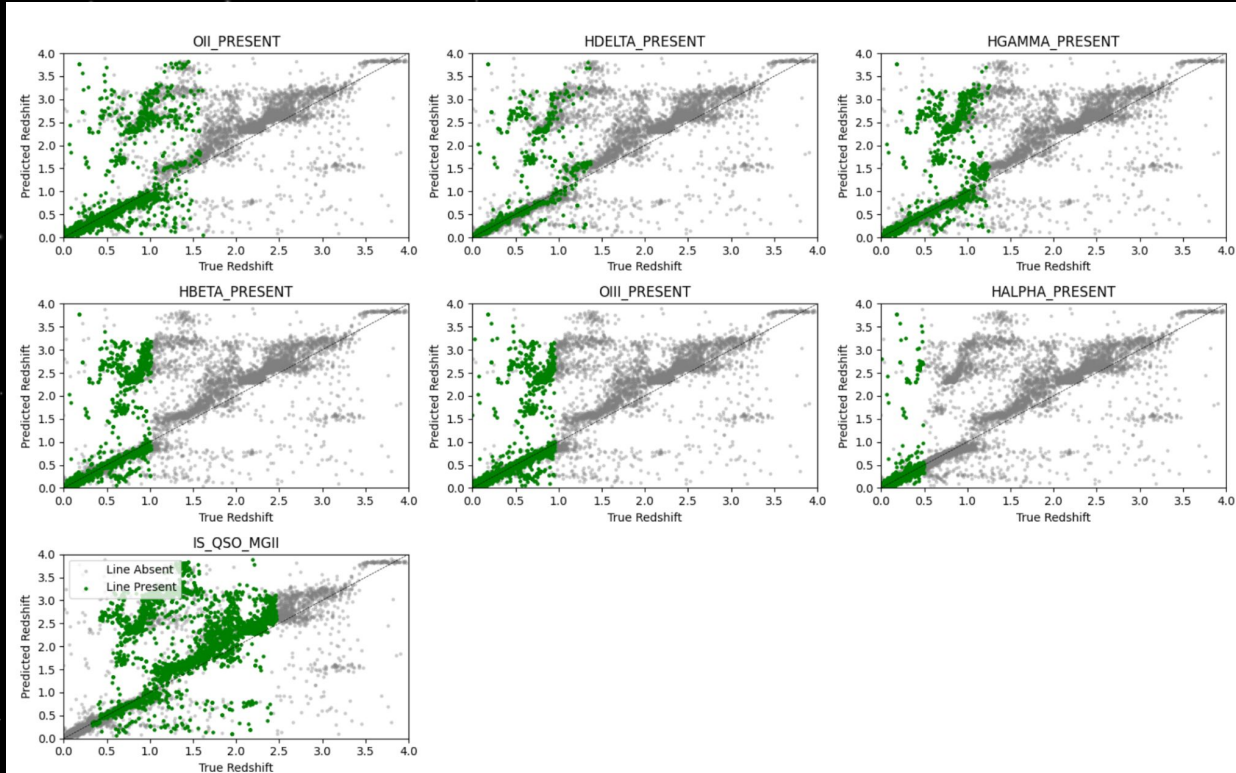


Trains on a simple reconstruction task to learn joint modality representation



Pre-trained model can then be applied to actual task, like predicting redshift

Emission lines present for sources

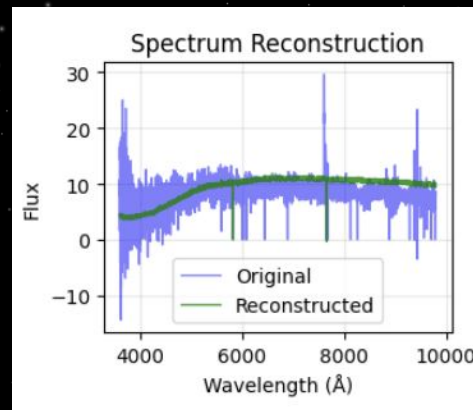
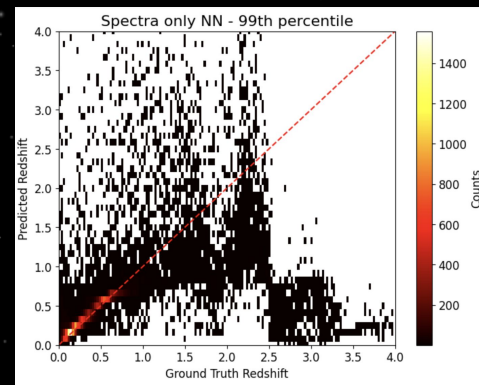
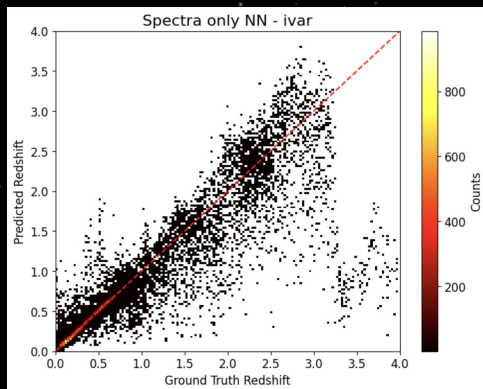


Data Cuts

- Find matches between the HSC and DESI datasets
 - Objects within 0.5 arcseconds of each other in the sky
- Duplicates and unreliable (high error) data points discarded
- Spectra with > 80% missing data and large spikes removed
- Camera coadd the spectra
 - Determine where red, blue, and green cameras overlap to get one uniform spectrum
- Processed dataset of 135,005 image-spectra observations, matched by target ID
- Determining ways to get rid of noisy data
 - Downsampling

Issues with Data

- Wide range of spectral flux values required normalization for the model to learn better
- Noisy spectrum data made it difficult for model to learn important spectral features



LSST Requirements

RMSE < 0.2

Outlier Rate $< .15$

Bias < 0.003

